Lecture 3: What Is Regression?

Goal: Introduce regression as estimating $E[Y|X]$; compare nonparametric and parametric approaches.

- What is regression?
  - Conditioning on predictors
  - Modeling conditional mean $E[Y|X]$
  - Prediction versus causal uses
  - Estimator: $\widehat{E}[Y|X]$
- Nonparametric regression: discrete X
  - Group means by education levels
  - Estimate $E[Y|X = x]$ within categories
- Nonparametric regression: continuous X
  - Kernel regression idea
  - Uniform kernel local averages
  - Bandwidth controls smoothing
  - Weighted kernels (Epanechnikov & Gaussian)
  - Bias–variance tradeoff
- Linear regression
  - Parametric form $E[Y|X] = \beta_0 + \beta_1 X$
  - Interpret intercept and slope
  - Justifications and limitations
- Local(ly) linear regressions
  - LOWESS/LOESS local linear smoothing
  - Bandwidth and weights in practice
  - GAM as flexible alternative